Securing Neural Interfaces: Architecture, Threat Taxonomy, and Neural Impact Scoring for Brain-Computer Interfaces

Kevin L. Qi
Qinnovate
qinnovate.com

October 2026 (v1.6; first posted February 2026)

Abstract

Brain-computer interfaces (BCIs) are moving from research tools toward commercially deployed medical devices, with companies including Neuralink, Synchron, Blackrock Neurotech, and Paradromics advancing toward regulatory approval and new entrants such as Merge Labs raising $252M in seed funding [4]. These devices read and, increasingly, write neural signals. We are unaware of a published security framework that scores the harms specific to such devices: the Common Vulnerability Scoring System (CVSS v4.0) has no metric for tissue damage, for unauthorised reading or disruption of neural function, for violated consent, or for effects that cannot be reversed.

We present a proposed, unvalidated framework with four parts: (1) an **addressing model** that tags what a device touches with eleven unordered codes for neural structures, the electrode-tissue interface and device systems; (2) **TARA**, a catalogue of BCI attack techniques, each with a tactic, an evidence status and a therapeutic analog where one exists; (3) **NISS**, an impact-only CVSS v4.0 extension with six neural-specific metrics; and (4) the **Neural Impact Chain**, which associates techniques with DSM-5-TR and ICD-10-CM diagnostic category references for threat modeling, not as diagnostic claims.

The analysis is of a frozen snapshot of 109 techniques (registry v1.5). In that snapshot CVSS cannot express the primary impact of 59 techniques (54.1%) and misses part of it for 30 more. Fifty-two techniques are in the direct-association class across 29 diagnostic category codes, and 78 (71.6%) have a recorded therapeutic analog: many attack mechanisms are shared with therapies, and what separates the two is consent, dosage and oversight. This revision (v1.6) replaces the earlier hourglass stack with addresses, adds an evidence rule and three scoring states, states NISS's limits, corrects case studies that did not match the registry, and argues why neurosecurity and neuroethics are needed now. Live registry figures (v4.3, 173 techniques) are reported beside the snapshot, not substituted into it. The framework, catalogue and scoring data are released under the Apache 2.0 license.

Keywords: brain-computer interface, neurosecurity, CVSS, threat taxonomy, neural scoring, neuroethics, DSM-5-TR, dual-use

Contents

1 Introduction **4**
1.1 Why Neurosecurity 4
1.2 The Scoring Gap 5
1.3 Contributions 5

2 Related Work **5**
2.1 Foundational Neurosecurity . . . 6
2.2 Empirical Attack Demonstrations . . . 6
2.3 Emerging Frameworks . . . 6
2.4 Neuroethics and Policy . . . 6
2.5 Cognitive Neurosecurity and Recent Surveys . . . 7
2.6 How QIF Differs . . . 7

3 From the Hourglass to an Addressing Model **7**
3.1 Why the Stack Was Withdrawn . . . 8
3.2 Addressing Version 2 . . . 8
3.3 Routes (Proposed) . . . 9

4 TARA: Threat Taxonomy **9**
4.1 Taxonomy Structure . . . 10
4.2 Evidence Classification . . . 10
4.3 Severity Distribution . . . 11
4.4 Category Breakdown . . . 11
4.5 Dual-Use Mapping . . . 12
4.6 Representative Techniques . . . 12

5 NISS: Neural Impact Scoring System **13**
5.1 Gap Analysis: Why CVSS Alone Is Insufficient . . . 13
5.2 Extension Metrics . . . 14
5.2.1 BI: Biological Impact . . . 14
5.2.2 CR: Cognitive Reconnaissance . . . 14
5.2.3 CD: Cognitive/Functional Disruption . . . 14
5.2.4 CV: Consent Violation . . . 15
5.2.5 RV: Reversibility . . . 15
5.2.6 NP: Neuroplasticity . . . 15
5.3 PINS Flag . . . 15
5.4 Scoring Formula . . . 16
5.5 Vector Format . . . 16
5.6 Scoring States and Stated Limits . . . 16
5.7 NISS Severity Distribution . . . 17
5.8 Neurological Extension (v1.1) . . . 17

6 Neural Impact Chain **18**
6.1 Pipeline Architecture . . . 18
6.2 NISS-to-DSM Bridge . . . 18
6.3 Coverage Statistics . . . 18
6.4 Diagnostic Cluster Distribution . . . 18
6.5 Association Classes . . . 19
6.6 Example Walkthrough . . . 19
6.7 Impact Chains as Associations (Live Dataset) . . . 19

7 Governance and Neuroethics **19**
7.1 Why Neuroethics . . . 20
7.2 How Neuroethics Enters the Score . . . 20
7.3 Consent Tiers . . . 20
7.4 What an Institutional Programme Has, and QIF Lacks . . . 21
7.5 Alignment with Policy Instruments . . . 21

7.6 Regulatory Mapping 22

8 Case Studies **22**
8.1 Case 1: Signal Injection (QIF-T0001, Gap Group 3) 22
8.2 Case 2: P300 Interrogation (QIF-T0035, Read-Only) 23
8.3 Case 3: Calibration Poisoning (QIF-T0058, Gap Group 2) 23
8.4 Case 4: Transducer Inversion (QIF-T0072, Gap Group 1) 23
8.5 Case 5: Real-World Vulnerability Disclosure 23

9 Limitations and Future Work **23**
9.1 No Empirical Validation on Real BCI Devices 23
9.2 Reproducibility of the v1.5 Figures 24
9.3 Diagnostic Category Associations Not Clinically Validated 24
9.4 NISS Weights Not Calibrated 24
9.5 No Interrater Reliability Study 24
9.6 Evidence and Completeness 24
9.7 Single-Author Bias 25
9.8 AI Tool Disclosure 25
9.9 Future Work 26

10 Conclusion **26**

1 Introduction

Brain-computer interfaces (BCIs) are no longer confined to research laboratories. Neuralink has implanted its N1 device in human patients [32], Synchron's Stentrode has demonstrated motor neuroprosthesis via neurointerventional surgery [35], and Blackrock Neurotech's Utah array has enabled high-performance speech neuroprostheses [52]. Paradromics is advancing toward high-bandwidth cortical interfaces for paralyzed patients. Investment in neurotechnology grew 700% between 2014 and 2021, with the global market projected to reach $25 billion by 2030 [48].

Regulation has begun to require cybersecurity for these devices without saying how to assess it. Section 3305 of the Food and Drug Omnibus Reform Act of 2022 (FDORA, Pub. L. 117-328) added Section 524B to the FD&C Act [50], mandating cybersecurity documentation—including threat modeling, software bills of materials, and vulnerability disclosure—in premarket submissions for connected medical devices, and since October 2023 the FDA has applied a Refuse-to-Accept policy to non-compliant submissions. Section 524B does not specify a threat taxonomy or scoring system, and we are unaware of provisions in the standards it draws on (CVSS, IEC 62443, AAMI TIR57, ISO 14971) for the risks specific to devices that read and write neural signals. We found no neural-specific security requirements in the EU Medical Device Regulation [15].

1.1 Why Neurosecurity

Four properties of neural devices, each documented in the literature, separate their security from that of other connected medical devices.

The device acts on a person. A stimulating device delivers current to tissue. The same mechanisms underlie established therapies such as deep brain stimulation [27] and transcranial magnetic stimulation [18], so an unauthorised change to stimulation parameters is a change to a treatment, not to a record. Some such effects may not be reversible, and a security assessment has to say so.

Neural data cannot be reissued. A leaked password can be changed; a recording of a person's brain activity cannot. Present devices do not read thoughts [24, 51]: speech neuroprostheses decode attempted speech using models trained on a cooperating user [52]. Information can nevertheless leak. Martinovic et al. [29] showed that P300 responses recorded with a consumer headset reveal information about PINs, banks and other private details at better than chance. If decoders keep improving, a recording retained today becomes a readout risk later; the catalogue records this as harvest-now-decode-later (QIF-T0166), with theoretical status.

The device can outlast its support. An implant stays in a person after the party that maintains it withdraws. In a modified Delphi study with a 24-expert panel, the 14 experts who completed the final survey agreed that responsibility for implanted neural devices after a trial needs explicit rules [20]. No attacker is needed for harm here, which is why this revision keeps such conditions in a separate hazards dataset (Section 4).

The attack surface spans biology and silicon. A recent review treats every element of the loop between brain, device, linked devices and servers, and every link between them, as an attack surface, and describes defence frameworks built specifically for BCIs as research prototypes [3].

Two consequences follow. First, an assessment needs dimensions that information-technology scoring lacks (Section 5). Second, the evidence base is thin: of the 173 techniques in the live catalogue, 95 are theoretical, and none of the catalogue's status values means that an attack has been carried out on a person with a device. Neurosecurity at this stage is therefore anticipatory

work, and its credibility depends on separating what has been shown from what is supposed. The evidence rules added in this revision (Section 4) exist for that reason.

1.2 The Scoring Gap

The Common Vulnerability Scoring System (CVSS v4.0) [16] is the industry standard for assessing vulnerability severity. CVSS captures exploitability characteristics—attack vector, complexity, privileges required, user interaction—and impact on the confidentiality, integrity and availability of information systems. It has no metric for:

- **Biological tissue damage**—seizure induction, tissue injury, involuntary motor activation
- **Unauthorised reading or disruption of neural function**—inference from neural data, perceptual or motor disruption
- **Consent boundaries**—the distinction between operating within consented parameters and covert manipulation
- **Damage reversibility**—IT assets can be restored from backup; neural tissue cannot
- **Neuroplastic consequences**—prolonged stimulation can cause lasting change to neural pathways

When a vulnerability in a BCI could induce seizures, expose information inferred from neural signals, or cause lasting harm, a CVSS base score alone is insufficient. In the 109-technique snapshot analysed here, CVSS cannot express the primary impact of 59 techniques and misses part of it for 30 more (Section 5).

1.3 Contributions

This paper presents a proposed, unvalidated framework with four parts:

1. **An addressing model** (Section 3): eleven codes in use—seven for neural structures (N1–N7), one for the electrode-tissue interface (I0) and three for device systems (S1–S3)—each with a membership rule. The codes are addresses with no order. Earlier versions of this paper presented them as an hourglass stack with a chokepoint; Section 3 says why that was withdrawn.
2. **TARA** (Section 4): a catalogue of 109 techniques across 15 tactics in the analysed snapshot (173 in the live registry), each classified by evidence status, severity and dual-use therapeutic potential. TARA is an independent taxonomy inspired by the structure of MITRE ATT&CK®.
3. **NISS** (Section 5): the Neural Impact Scoring System, an impact-only CVSS v4.0 extension designed to conform with FIRST.org's extension mechanism (User Guide §3.11) [17], adding six neural-specific metrics, with stated limits and three scoring states.
4. **The Neural Impact Chain** (Section 6): a threat-modeling method that associates techniques with diagnostic category references through addresses, structures and functions to DSM-5-TR and ICD-10-CM codes [1]. The codes are category references, not diagnostic claims.

Section 7 sets out why a security framework needs neuroethics and how it enters the scoring; Section 8 presents case studies, and Section 9 states limitations and what this revision changed. The framework, catalogue and scoring system are released under the Apache 2.0 license.

2 Related Work

Research at the intersection of cybersecurity and neurotechnology has progressed through three phases: foundational framing, empirical demonstration, and emerging policy response. QIF builds on each while addressing gaps left by prior work.

2.1 Foundational Neurosecurity

Denning, Matsuoka, and Kohno [12] introduced the term "neurosecurity" and analyzed attack surfaces in implantable neurostimulators. Their work identified wireless reprogramming, battery depletion, and signal injection as threat categories, but predated modern high-bandwidth BCIs and did not propose a scoring system.

Bonaci, Calo, and Chizeck [6] extended this with "App Stores for the Brain," examining privacy threats from third-party BCI applications. They identified the need for access control at the neural data layer—a concern our governance framework addresses through consent tiers (Section 7).

2.2 Empirical Attack Demonstrations

Martinovic et al. [29] showed at USENIX Security 2012 that P300 event-related potentials recorded with consumer-grade EEG headsets leak information about PINs, banks and other private details at better than chance. TARA (Section 4) records this as P300 interrogation (QIF-T0035), with status demonstrated.

Ienca and Haselager [23] broadened the scope to "hacking the brain," analyzing BCI security through the lens of informational and physical integrity. They proposed that BCI security requires both traditional cybersecurity measures and novel neuroethical safeguards—a dual requirement that QIF implements through the parallel CVSS/NISS scoring architecture.

Meng et al. [30] established an adversarial robustness benchmark for EEG-based BCIs, evaluating multiple adversarial defense approaches across neural network architectures and EEG datasets. Their work focuses on machine learning adversarial robustness; TARA extends coverage to physical-layer, protocol-layer, and cognitive-layer threats.

2.3 Emerging Frameworks

Schroder et al. [43] (2025) developed a threat model for network-based attacks on next-generation BCIs and identifying associated regulatory changes needed to address them. Their work contributes threat categorization across network pathways but does not propose a scoring system, formal threat taxonomy with technique-level granularity, or clinical impact mapping.

Camara et al. [9] and Rushanan et al. [42] surveyed security and privacy in implantable medical devices more broadly, covering pacemakers, insulin pumps, and cochlear implants alongside neurostimulators. Halperin et al. [19] demonstrated practical wireless attacks against pacemakers, establishing that implanted medical device security is not theoretical. These works address the broader medical device landscape; QIF focuses specifically on the unique challenges of bidirectional neural interfaces.

2.4 Neuroethics and Policy

Ienca and Andorno [22] proposed four fundamental neurorights: cognitive liberty, mental privacy, mental integrity, and psychological continuity. Yuste et al. [53] added equal access and protection from algorithmic bias. These philosophical frameworks establish *what* must be protected; QIF provides the technical architecture for *how*.

Muñoz et al. [33] and Lázaro-Muñoz et al. [28] contributed empirical grounding through researcher interviews, documenting clinician concerns about adaptive DBS ethics (72% cited uncertainty about risks) and post-trial access obligations for implanted neural devices.

UNESCO's 2025 Recommendation on the Ethics of Neurotechnology [48]—adopted by 194 Member States—is a global normative instrument for neurotechnology governance. The OECD had previously published a Recommendation on Responsible Innovation in Neurotechnology [34] covering 36 member countries. Chile amended its constitution in 2021 to require the law to

safeguard brain activity and the information derived from it [40]. These policy instruments establish principles and values; QIF demonstrates technical implementation (Section 7).

2.5 Cognitive Neurosecurity and Recent Surveys

Two recent preprints from one group approach the problem from the cognitive side. Bagley and Petritsch [2] attribute the gap between calls for neurotechnology security and applied work to the lack of a precise way to describe the problems, and propose *Cognitive Neurosecurity*: cognitive state variables combined into a high-dimensional vector, with two attack classes, readout attacks that identify cognitive states against a person's will and alteration attacks that modify them. They state that the formalism borrowed from quantum theory implies no physically quantum basis for cognition, treat defence as optimisation, expect defence against alteration to depend on detection, and argue that open publication of attack structures is the most ethical option. Their readout/alteration split corresponds to the NISS split between Cognitive Reconnaissance and Cognitive Disruption (Section 5). They also note that small alterations spread across many people could stay below individual-level detection; TARA records techniques one device and one person at a time, so this is a gap in its model.

Bagley et al. [3] review threat vectors and defences across the loop from brain to servers. Mapping each vector in that review against TARA found four without a corresponding technique; they were added as QIF-T0166, T0170, T0171 and T0177, and one finding was filed as a variant (QIF-T0024-V01). Four of the five rest only on preprint evidence and are unscored; QIF-T0166 has one indirect peer-reviewed source and carries a provisional range. Both works are preprints, the mappings were made with AI assistance, and their authors have not reviewed them.

2.6 How QIF Differs

Prior work has addressed BCI security, neuroethics and vulnerability scoring largely separately. We are unaware of prior work that combines the following four elements in one open dataset:

1. **Addressing**: codes with membership rules for what a device touches
2. **Taxonomy**: technique-level entries with an evidence status and an evidence rule (vs. category-level surveys)
3. **Scoring**: a CVSS-compatible, impact-only extension with neural-specific metrics and explicit scoring states
4. **Clinical association**: links from techniques to diagnostic category references for threat modeling

We found no prior work that systematically associates vulnerability techniques with DSM-5-TR diagnostic category codes, but our search was not exhaustive, and the association method (Section 6) has not been clinically validated.

3 From the Hourglass to an Addressing Model

Versions up to v1.5 of this paper presented an 11-band *hourglass*: seven neural bands (N7 neocortex down to N1 spinal cord), one interface band (I0, the electrode-tissue boundary) and three synthetic bands (S1–S3), read as a stack. The model borrowed from the OSI reference model [55] and from the Internet's narrow waist [11], and made three claims: that I0 is a natural security chokepoint through which all signals pass; that every component of a BCI maps to exactly one band, so that no attack surface falls outside the model; and that behaviour becomes less deterministic ascending the neural bands. This revision withdraws all three and replaces the stack with addresses. This section records what changed and why.

3.1 Why the Stack Was Withdrawn

The numbering carries no flow. If the bands were a stack, neural pathways should mostly join numerically adjacent bands. Recomputing each pathway's origin and target bands from the structure table gives 65 links between distinct N codes, of which 18 (27.7%) join adjacent codes. Over all 5,040 possible numberings of the seven codes the mean is 28.6%, and 60% of numberings do at least as well as the published one. The order is no better than chance.

The thalamus is not a gate. The v1.5 text used thalamocortical relay [45] to support signal-plausibility checks, and earlier project documents described the thalamus as the brain's firewall. The thalamus is the principal route by which sensory input, cerebellar output and basal ganglia output reach neocortex, and transmission through it is state-dependent [44]. It is not a security control: brainstem neuromodulatory systems also reach cortex by extra-thalamic routes [13], motor thalamus integrates cortical, cerebellar and basal ganglia input [7], and olfaction is not thalamus-free, since piriform cortex projects to the mediodorsal thalamus [10].

I0 is a boundary, not a chokepoint. The electrode-tissue boundary is where a recording or stimulating device transduces between ionic and electronic current, and it remains an address. It is not a point through which all attacks pass: of the 173 techniques in the live catalogue, 110 carry no I0 tag and 53 are tagged only at device codes, among them consumer-sensor side channels that never touch an electrode.

The model was not complete. Nerves, roots, ganglia and sense organs had no address, so the completeness claim did not hold. A periphery code (PER) is proposed for them and is not yet integrated.

Determinacy was untested. The per-band determinacy classes and "QI" ranges had no defined estimator. They are now recorded as untested hypotheses (`datalake/qif-hypotheses.json`), proposed for the earlier band memberships.

3.2 Addressing Version 2

The seven N codes are retained as identifiers. Each structure has one address, set by the adult territory that holds its principal neurons; circuits such as the basal ganglia and the limbic system are functional tags a structure can carry, not addresses. Table 1 gives the codes in use. **The number carries no severity, processing or traversal meaning.**

Table 1: Address codes in use (addressing version 2). Listed by code, not by rank.

Code	Name	Membership rule
N7	Neocortex	Isocortex, including agranular motor areas
N6	Allocortex	Allocortex, mesocortex and pallial nuclei, including basolateral amygdala (v1.5: "Limbic System")
N5	Cerebral nuclei	Subpallial nuclei: striatum, pallidum, basal forebrain, central and medial amygdala (v1.5: "Basal Ganglia")
N4	Diencephalon	Thalamus, subthalamus, hypothalamus, epithalamus
N3	Cerebellum	Cerebellar cortex and nuclei
N2	Brainstem	Midbrain, pons and medulla, including substantia nigra and ventral tegmental area
N1	Spinal cord	Spinal cord parenchyma; roots and ganglia are outside it
I0	Neural interface	Electrode-tissue boundary where biological signals become electronic
S1	Analog front-end	Amplification, filtering, ADC/DAC conversion
S2	Digital processing	Decoding algorithms, classification, digital signal processing
S3	Application	Clinical software, user interface, data storage, network

Three structures changed address. The substantia nigra and ventral tegmental area moved from N5 to N2, because they lie in the midbrain, and the subthalamic nucleus moved from N5 to N4. All three keep their basal ganglia or reward tags. Subthalamic deep brain stimulation, which v1.5 placed at N5, is therefore an N4 example. Because codes keep their identifiers while membership changes, the atlas carries `addressing_version: 2`, which should be cited with any band reference. Pathway bands are now computed from the structure table by a script with a check mode, which the build does not yet run.

Fourteen statements in project documents drew security conclusions from the gate analogy, chiefly that a deeper implant bypasses fewer protective layers. They are marked under review, neither deleted nor asserted. The NISS formula has no term for implant depth: all 129 stored scores reproduce exactly from their six metric values, although that does not show whether depth reasoning influenced the choice of any metric level.

3.3 Routes (Proposed)

Signal flow, which the stack could not carry, is moved to a proposed route view, staged and not integrated. A route is an ordered list of synaptic hops, each with its sign, transmitter and own evidence. A source check of all 90 hops in the 17 staged routes found 47 supported by a source a checker read, 42 partly supported and 1 contradicted for primates; 3 hops are recorded as demonstrated in humans and 85 as inferred by homology from other species. The routes are suitable for tagging what a device touches and for teaching. They are not a basis for claims about what an attack can reach in a person.

Effect on this paper. The band annotations in the snapshot analysis (Sections 4–6) use the version 1 memberships. Those analyses are not recomputed here; a reader using a band count from them should treat it as a version 1 figure.

4 TARA: Threat Taxonomy

The Therapeutic Applications & Risk Assessment (TARA) registry catalogues BCI attack techniques with a structure inspired by MITRE ATT&CK [31], independently developed to cover neural, cognitive, and physical safety domains. Each technique is simultaneously an attack vector, an ethical risk, and—where applicable—a therapeutic application.

Why not ATT&CK directly? MITRE ATT&CK is designed for enterprise IT and mobile environments where adversary objectives center on data exfiltration, lateral movement, and persistence. BCI threats differ in three fundamental ways that make direct adoption inadequate: (1) the target is biological tissue, not an information system—attacks can cause seizures, tissue necrosis, or cognitive coercion, none of which map to ATT&CK's impact categories; (2) the same technique is built on the same mechanism as a therapy (e.g., the stimulation used in deep brain stimulation for Parkinson's disease can also be delivered without authorisation), requiring a dual-use classification that ATT&CK has no mechanism to express; and (3) the attack lifecycle must span from radio-frequency transmission through silicon processing to neural tissue, crossing the bio-digital boundary that ATT&CK's purely digital model does not address. TARA adopts ATT&CK's proven structural methodology—techniques organized into tactics and domains—while building an independent taxonomy purpose-built for the BCI threat landscape.

4.1 Taxonomy Structure

The TARA registry organizes 109 techniques into 15 tactics across 8 operational domains. Technique identifiers follow the format `QIF-TXXX`, where the numeric suffix provides sequential ordering.

Domains. The eight domains span the full attack lifecycle:
1. **Neural (N)** — Direct interaction with neural tissue (scan, injection, manipulation)
2. **BCI System (B)** — System-level intrusion and evasion
3. **Protocol (P)** — Protocol disruption and communication attacks
4. **Data (D)** — Data harvesting and exfiltration
5. **Cognitive (C)** — Cognitive exploitation and imprinting
6. **Countermeasure (M)** — Surveillance and monitoring
7. **Evasion (E)** — Defense evasion and anti-detection
8. **Sensor (S)** — Consumer device side-channel attacks

Tactics. The 15 tactics follow a lifecycle structure analogous to ATT&CK, adapted for BCI operations: Neural Scan, BCI Intrusion, Neural Injection, Cognitive Imprinting, BCI Evasion, Data Harvesting, Neural Manipulation, Evasion/Rootkit Deployment, Monitoring/Surveillance, Protocol Disruption, Cognitive Exploitation, Signal Replay, Signal Harvesting, Signal Fingerprinting, and Signal Chaining.

4.2 Evidence Classification

Each technique carries an evidence status. Table 2 gives the snapshot counts used in this paper beside the live counts.

Table 2: TARA evidence status: the v1.5 snapshot analysed here and the live registry (v4.3, read 2026-10-17).

Status	Definition (live registry)	v1.5	Live
Confirmed	Retired 2026-10-16 (see text)	19	—
Demonstrated	A published source executes the technique against a real system	35	48
Emerging	Shown against an analogous system or in part	22	27
Theoretical	The mechanism is established but no attack using it has been shown	31	95
Plausible	The mechanism is contested or indirect; the attack is reasoned	1	2
Speculative	Neither mechanism nor attack is established	1	1
	Total	109	173

CONFIRMED retired. The status *confirmed* was retired on 2026-10-16 after an audit found that it spanned unrelated kinds of evidence; for example, it had been given to signal injection (QIF-T0001) on the strength of a position paper and a P300 probing study, neither of which demonstrates signal injection. The 27 entries that carried it in the live registry were reclassified as theoretical (21), demonstrated (5) and plausible (1), each keeping its prior value and a reason. The 19 *confirmed* entries of the snapshot should be read with this in mind. **No status value means that an attack has been carried out on a person with an implanted or worn device.**

Evidence rule. Every live entry records whether its sources are peer-reviewed, preprint or absent (158, 7 and 8 entries). **A technique with no peer-reviewed source may not hold a status above theoretical and does not carry a NISS score.** Papers that test a defence are held in a separate field, so they cannot satisfy the citation requirement for an attack. When the rule was applied, 34 entries broke it; 24 were given a verified peer-reviewed source and 10 lost their scores. "Peer-reviewed" currently means that no preprint identifier was found, which is weaker than verified.

Hazards held separately. Some conditions harm a device user with no attacker, such as a manufacturer withdrawing support for an implant. They do not belong in a catalogue of adversary actions, so 23 entries are held in `datalake/qif-hazards.json`. One, QIF-H0001 (loss of manufacturer or sponsor support), rests on two peer-reviewed sources [20, 54] and is theoretical. The other 22 are unsourced pointers to catalogue records that describe a therapy or physical mechanism; they are not warnings about those therapies and assert nothing about incidence in people.

4.3 Severity Distribution

Each technique also carries a filed severity rating. In v1.5 this table was labelled as CVSS v4.0 base ratings; it is in fact the registry's `severity` field, assigned by hand and not computed from the CVSS vectors. In the live registry it agrees with the NISS severity for 30 of 129 scored techniques, so it should not be read as either a CVSS or a NISS result.

Table 3: TARA filed severity distribution (v1.5 snapshot; hand-assigned).

Severity	Count	Percentage
Critical	29	26.6%
High	60	55.0%
Medium	17	15.6%
Low	3	2.8%

The weighting toward high and critical reflects the author's judgement at filing, not a computed result.

4.4 Category Breakdown

The 109 techniques distribute across eight operational categories:

Table 4: TARA techniques by operational category.

ID	Category	Count
SE	Signal Eavesdropping	20
CI	Cognitive Interference	18
EX	Data Exfiltration	19
DM	Data Manipulation	15
SI	Signal Injection	13
PE	Privilege Escalation	9
DS	Denial of Service	8
PS	Physical Safety	7

4.5 Dual-Use Mapping

TARA assesses each technique for therapeutic analogs. The same physical mechanisms that enable attacks—electromagnetic stimulation, signal decoding, neuromodulation—are the mechanisms underlying established therapies such as deep brain stimulation (DBS) [27], transcranial magnetic stimulation (TMS) [18], and neurofeedback.

Table 5: Dual-use classification of 109 TARA techniques.

Classification	Definition	Count	%
Confirmed	Published clinical use exists	53	48.6%
Probable	Under active clinical investigation	16	14.7%
Possible	Theoretical therapeutic mapping	9	8.3%
Silicon Only	No tissue analog; purely digital	25	22.9%
Not Mapped	Dual-use assessment pending	6	5.5%

Of the 109 techniques, 78 (71.6%) have a confirmed, probable or possible therapeutic analog, and 25 are silicon-only. The finding is that many attack mechanisms are shared with therapies, not that every attack has a therapeutic counterpart: the mechanism is shared, and the boundary between therapy and attack is consent, dosage and oversight. Defences cannot simply remove a capability that treatment depends on.

4.6 Representative Techniques

Table 6 shows five registry entries spanning categories and dual-use classes. In v1.5 this table listed descriptions and identifiers that did not match the registry entries carrying those identifiers (for example, QIF-T0015 was described as a P300 side channel but is directed-energy thermal damage); it now quotes the registry.

Table 6: Five representative TARA entries as stored in the registry (status: snapshot → live).

ID	Name	Status	Dual-use	Therapeutic analog
T0001	Signal injection	Confirmed → Theoretical	Confirmed	tDCS/tACS neuromodulation
T0035	P300 interrogation	Demonstrated	Confirmed	P300 BCI for locked-in patients
T0058	Calibration poisoning	Demonstrated	Probable	Adaptive BCI calibration
T0072	Transducer inversion (speaker-to-microphone)	Confirmed → Demonstrated	Silicon only	—
T0090	WiFi CSI passive body sensing	Demonstrated	Confirmed	Contactless vital-sign monitoring

5 NISS: Neural Impact Scoring System

The Neural Impact Scoring System (NISS) is a proposed CVSS v4.0 extension designed to conform with FIRST.org's official extension mechanism (User Guide §3.11) [17]. NISS adds six metrics that capture dimensions CVSS was never designed to express: biological tissue damage, cognitive reconnaissance, cognitive disruption, consent boundary violations, damage reversibility, and neuroplastic consequences.

NISS scores impact only; it has no likelihood or exploitability term and is meant to sit beside a CVSS base score, not to replace it. It is also the technique-level input to a proposed multi-layer Neurosecurity Score (NSv2.1), which is unvalidated and outside the scope of this paper.

5.1 Gap Analysis: Why CVSS Alone Is Insufficient

Each TARA technique carries a CVSS v4.0 base vector and a gap group recording how much of its impact CVSS fails to capture. Table 7 gives the counts published in v1.5 beside those recomputed from the registry file committed with v1.5.

Table 7: CVSS v4.0 gap groups across the 109 snapshot techniques: as published in v1.5, and recomputed from the committed snapshot file.

Group	Gap Description	v1.5 text	Snapshot file
1	CVSS captures most impact; NISS adds nuance	13	14
2	CVSS captures exploitability but misses part of the impact	31	30
3	CVSS cannot express the primary impact	64	59
—	No gap group assigned	1	6
	Groups 1–3	**108 (99.1%)**	**103 (94.5%)**

The v1.5 counts could not be reproduced from the registry file committed with that version, which gives 59 techniques (54.1%) in Group 3 and 103 (94.5%) in any group; this revision uses the reproducible counts in its text and retains the published ones in the table for the record. Group 1 techniques are mostly captured by CVSS, so the share that *needs* NISS is better read as Groups 2 and 3, 89 techniques (81.7%). In Group 3 the most severe consequence is tissue damage, disruption of neural function or lasting harm—dimensions for which CVSS has no metric. Live: 103 of 173 (59.5%) carry a gap group.

5.2 Extension Metrics

NISS defines six extension metrics, each with a graduated value set. The metrics are designed to be orthogonal to CVSS base metrics: they capture impact dimensions that exist only because the target system interfaces with biological neural tissue.

5.2.1 BI: Biological Impact

Direct harm to neural tissue, organs, or physiological function. This dimension has no equivalent in CVSS.

Value	Label	Score	Description
N	None	0.0	No tissue interaction or physical harm
L	Low	3.3	Temporary discomfort, minor sensory disruption, reversible tissue stress
H	High	6.7	Significant tissue damage, seizure induction, involuntary motor activation
C	Critical	10.0	Life-threatening or permanently disabling neural harm

5.2.2 CR: Cognitive Reconnaissance

Degree of unauthorised read access: decoding of speech or motor intent by models trained on a cooperating user, inference from neural data, intent extraction. CVSS treats such reads only as loss of data confidentiality. CR maps to the Cognitive Liberty and Mental Privacy neurorights dimensions.

Value	Label	Score	Description
N	None	0.0	No cognitive read access
L	Low	3.3	Partial intent inference, coarse emotional state detection
H	High	6.7	Decoding of intent, identity inference, or detailed neural data extraction
C	Critical	10.0	Sustained, broad inference from neural data in real time

5.2.3 CD: Cognitive/Functional Disruption

Degree of unauthorized disruption to cognitive processing, sensory perception, motor output, or autonomic regulation—including perception manipulation, identity modification, cognitive coercion, and false stimuli injection. CD maps to the Mental Integrity neurorights dimension and extends to ICD-10-CM chapters G (nervous system), H (eye/ear), and R (symptoms/signs).

Value	Label	Score	Description
N	None	0.0	No cognitive write access or alteration
L	Low	3.3	Minor perceptual distortion, detectable interference
H	High	6.7	Perception manipulation, involuntary behavioral changes
C	Critical	10.0	Cognitive coercion, identity modification, or complete loss of cognitive autonomy

5.2.4 CV: Consent Violation

Degree of violation of informed consent or cognitive autonomy. Ordered by severity: covert (implicit) violations are worse than detectable (explicit) ones.

Value	Label	Score	Description
N	None	0.0	Operating within explicitly consented boundaries
P	Partial	3.3	Action exceeds scope but subject retains some awareness
E	Explicit	6.7	Direct violation of consent boundaries, but detectable
I	Implicit (covert)	10.0	Covert manipulation the patient cannot detect or refuse

5.2.5 RV: Reversibility

Whether the damage can be undone. IT assets can be restored from backup. Neural tissue cannot be rebooted.

Value	Label	Score	Description
F	Full	0.0	Effects fully reverse when attack stops
T	Temporary	3.3	Effects reverse over hours to days
P	Partial	6.7	Some effects permanent, some reversible
I	Irreversible	10.0	Permanent neural tissue destruction or cognitive change

5.2.6 NP: Neuroplasticity

Whether the attack exploits or induces neuroplastic changes—the brain's ability to rewire itself. This has no digital equivalent.

Value	Label	Score	Description
N	None	0.0	No neuroplastic effect
T	Temporary	3.3	Short-term synaptic changes that decay within hours–weeks
P	Partial	6.7	Moderate synaptic reorganization persisting weeks to months
S	Structural	10.0	Long-term or permanent neural pathway changes

5.3 PINS Flag

NISS introduces the Potential Impact to Neural Safety (PINS) flag—a binary indicator triggered when:

$$\text{PINS} = \begin{cases} \texttt{true} & \text{if BI} \geq \text{High} \ \vee \ \text{RV} = \text{Irreversible} \\ \texttt{false} & \text{otherwise} \end{cases} \tag{1}$$

A PINS flag calls for safety review regardless of overall score. In the snapshot, 33 of 109 techniques are PINS-flagged (30.3%); in the live registry, 37 of the 129 scored techniques are.

5.4 Scoring Formula

The NISS score is computed as the weighted mean of the six metric scores, rounded up to one decimal place:

$$\text{NISS} = \frac{w_{\text{BI}} \cdot \text{BI} + w_{\text{CR}} \cdot \text{CR} + w_{\text{CD}} \cdot \text{CD} + w_{\text{CV}} \cdot \text{CV} + w_{\text{RV}} \cdot \text{RV} + w_{\text{NP}} \cdot \text{NP}}{w_{\text{BI}} + w_{\text{CR}} + w_{\text{CD}} + w_{\text{CV}} + w_{\text{RV}} + w_{\text{NP}}} \tag{2}$$

In the default profile, the weights are $w_{\text{BI}} = w_{\text{CV}} = w_{\text{RV}} = w_{\text{NP}} = 1.0$ and $w_{\text{CR}} = w_{\text{CD}} = 0.5$. This normalization ensures that the split of the former Cognitive Integrity (CG) metric into Cognitive Reconnaissance (CR) and Cognitive Disruption (CD) in v1.1 does not inflate the cognitive dimension's influence: at $0.5 + 0.5 = 1.0$ out of a total weight sum of 5.0, cognition retains its original 20% share from the five-metric v1.0 formula. The CR/CD split reflects the fundamental distinction between read attacks (inference from neural data) and write attacks (perception manipulation, cognitive coercion)—attack vectors with different mitigations that map to different neurorights dimensions. NISS supports four context profiles with differential weights that override the default normalization:

- **Clinical**: BI, CD and RV at 2.0, others at 1.0
- **Research**: CR and CV at 2.0, CD and NP at 1.5
- **Consumer**: CR and CV at 2.0, others at 1.0
- **Military**: BI, CR and CD at 2.0, RV and NP at 1.5, CV at 0.5

All stored scores use the default weights. The military profile halves the weight of consent, a choice with ethical content that has not been reviewed.

5.5 Vector Format

The NISS vector rides alongside the CVSS v4.0 base vector:

```
CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/
  VC:H/VI:H/VA:H/SC:N/SI:N/SA:N
NISS:1.1/BI:H/CR:H/CD:C/CV:I/RV:P/NP:S
```

This dual-vector architecture means security teams can triage using familiar CVSS scores while BCI-specific teams see the neural dimensions that determine whether a vulnerability is a software bug or a patient safety emergency.

5.6 Scoring States and Stated Limits

A NISS score is six judgements about impact. Scoring a technique whose sources support only one of them means inventing the other five, so the registry now distinguishes three *scoring states*:

- **Scored**: all six metrics are supported by the technique's sources (129 live techniques; these were scored before the rule was introduced and have not yet been re-examined against it).
- **Provisional**: some metrics are supported; the record stores those values, a source for each, and the lowest and highest score the formula could return (10). A range of 0.4 to 9.4 says that one metric is known and five are not; it is a bound, not an estimate, and not a severity.
- **Unscored**: no peer-reviewed source, or no metric supported (34).

Read the scoring state before the score. Three limits follow from the formula and the data:

1. **Impact only.** NISS has no likelihood term, so a technique with several disclosed vulnerabilities can score low.
2. **Read-only harm cannot exceed 3.0.** With CR at Critical, CV at Implicit and nothing physical, the score is $(0.5 \cdot 10 + 10)/5 = 3.0$. NISS therefore rates every pure privacy-and-consent harm as low, whatever its scale. This is a property of the default weights, and it is the clearest reason to report the vector, not only the number.
3. **No calibration or inter-rater study.** The levels were assigned by one author with AI assistance (Section 9).

NISS scores signal-level interference with a person's neural function and the consent under which it occurs; it does not measure mental states, and a score is not a clinical or legal finding. Whether NISS should be reported as a single number or as its vector with the PINS flag is an open question.

5.7 NISS Severity Distribution

The NISS severity distribution in the snapshot, as published in v1.5:

Table 8: NISS severity distribution (109 snapshot techniques, as published in v1.5).

NISS Severity	Count	%
High	19	17.4%
Medium	37	33.9%
Low	52	47.7%
None	1	0.9%

The registry file committed with v1.5 gives 21 high, 37 medium, 50 low and 1 none, and several of its stored scores do not reproduce from Equation 2; the live scores do. The NISS distribution sits lower than the filed severity (Table 3) because NISS separates purely digital impact from impact on a person.

5.8 Neurological Extension (v1.1)

NISS v1.0 mapped technique outcomes exclusively to DSM-5-TR psychiatric categories. This created a systematic blind spot: neurological disruptions—loss of smell, tinnitus, vestibular dysfunction, motor tremor, autonomic dysregulation—have no DSM-5-TR entry because the DSM is a psychiatric manual, not a neurological reference.

NISS v1.1 extends the impact chain to 42 neurological conditions drawn from ICD-10-CM chapters G (diseases of the nervous system), H (diseases of the eye and ear), and R (symptoms and signs not elsewhere classified), using Adams & Victor's *Principles of Neurology* (12th ed.) as the primary clinical reference. These conditions are organized into seven categories:

- **Sensory** (14 conditions): auditory (tinnitus, hearing loss), visual (scotoma, cortical blindness), vestibular (vertigo, nystagmus), olfactory (anosmia, parosmia), gustatory (ageusia, dysgeusia), and somatosensory (paresthesia, allodynia)
- **Motor** (8 conditions): tremor, dystonia, ataxia, spasticity, fasciculations, myoclonus, bradykinesia, dyskinesia
- **Autonomic** (6 conditions): orthostatic hypotension, hyperhidrosis, cardiac arrhythmia, thermoregulatory dysfunction, gastroparesis, neurogenic bladder
- **Consciousness** (4 conditions): syncope, absence seizures, altered sensorium, coma
- **Speech/Language** (4 conditions): expressive aphasia, receptive aphasia, dysarthria, alexia
- **Reflex** (3 conditions): hyperreflexia, areflexia, abnormal plantar response
- **Pain** (3 conditions): neuropathic pain, trigeminal neuralgia, central pain syndrome

Each condition maps to specific QIF hourglass bands, neural pathway dependencies, and NISS metrics. The CD metric was broadened from "Cognitive Disruption" to "Cognitive/Functional Disruption" to encompass sensory, motor, and autonomic disruption alongside cognitive interference. This definitional change preserves backward compatibility: all existing CD scores remain valid, and the scoring formula is unchanged. The extension adds coverage for neurological outcomes that were previously invisible to the scoring system.

The neurological conditions are queryable via KQL as the `neurological_conditions` table, enabling cross-correlation with TARA techniques, neural pathways, and hourglass bands.

6 Neural Impact Chain

The Neural Impact Chain (NIC) is a six-stage threat-modeling method that associates attack techniques with DSM-5-TR and ICD-10-CM diagnostic category references [1]. The codes are **diagnostic category references for threat modeling, not diagnostic claims**: an association says that a technique and a condition category share an address and a function, not that the technique causes the condition. The question the NIC helps a threat modeler ask is *which clinical categories should a review of this technique consider?* The mappings have not been reviewed by clinicians (Section 9).

6.1 Pipeline Architecture

The NIC traces each technique through six stages:

1. **Technique**: The TARA attack technique (e.g., cortical signal injection)
2. **Address**: The code(s) the technique is tagged at (e.g., N7 Neocortex, N6 Allocortex; version 1 memberships in the snapshot)
3. **Neural Structure**: The anatomical structure at risk (e.g., hippocampus, prefrontal cortex, amygdala)
4. **Cognitive Function**: The function that structure supports (e.g., memory consolidation, executive control, emotional regulation)
5. **NISS Score**: The neural impact score, particularly the BI, CR, CD, and NP metrics that correlate with clinical outcomes
6. **Category reference**: The DSM-5-TR/ICD-10-CM code(s) for the condition category most closely associated with disruption of that function

6.2 NISS-to-DSM Bridge

A rule assigns each technique to a DSM-5-TR category cluster according to which NISS metric dominates its profile. The rule is a modeling choice, not a clinical finding:

- **BI-driven** → Motor/Neurocognitive cluster: movement and neurocognitive categories
- **CR/CD-driven** → Cognitive/Psychotic cluster: perceptual and cognitive categories
- **CV-driven** → Mood/Trauma cluster: trauma- and stressor-related categories
- **NP/RV-driven** → Persistent/Personality cluster: categories of lasting change
- **No neural impact** → Non-diagnostic: Silicon-only techniques with no category association

6.3 Coverage Statistics

All 109 TARA techniques have been mapped through the NIC pipeline:

Table 9: Neural Impact Chain mapping results across 109 techniques.

Metric	Value
Techniques mapped	103 / 109 (94.5%)
Unique category codes	29
Diagnostic clusters	5
Direct association	52 (47.7%)
Indirect association	9 (8.3%)
No association (silicon-only)	42 (38.5%)

6.4 Diagnostic Cluster Distribution

The five diagnostic clusters and their technique counts:

Table 10: DSM-5-TR diagnostic category clusters (snapshot), for threat modeling.

Cluster	Count	Representative DSM-5-TR Codes
Non-diagnostic	42	— (silicon-only techniques)
Mood/Trauma	21	F43.10 (PTSD), F32.9 (MDD), F44.9
Cognitive/Psychotic	17	F06.0 (psychosis), R41.3 (cognitive decline)
Motor/Neurocognitive	16	G25.9 (movement disorder), G31.84
Persistent/Personality	7	F07.0 (personality change)

The Mood/Trauma cluster is the largest diagnostic category cluster (21 techniques), because the bridge rule sends consent-dominated techniques there. That is a consequence of the rule, not an observation about outcomes in people.

6.5 Association Classes

Each technique is placed in one of three association classes:

- **Direct** (52 techniques, 47.7%): the technique acts on the structure or function associated with the category. Example: unauthorised cortical stimulation is associated with seizure-related categories.
- **Indirect** (9 techniques, 8.3%): the association runs through a downstream effect. Example: exfiltration of neural data is associated with anxiety-related categories through the effect of the disclosure, not through the neural signal.
- **None** (42 techniques, 38.5%): Silicon-only techniques with no direct neural interaction and thus no category association.

6.6 Example Walkthrough

Consider **QIF-T0001, Signal injection**, as stored in the live registry. (The v1.5 walkthrough gave a vector, a score of 8.7 and three codes that did not match the registry entry; it is replaced here.)

1. **Technique**: QIF-T0001 (signal injection at the electrode-tissue interface)
2. **Address**: I0, N1
3. **Mechanism**: electrical current delivered at the interface, modulating local field potentials
4. **NISS**: `BI:H/CR:H/CD:H/CV:E/RV:P/NP:T` → 6.1 (Medium), PINS set because BI is High
5. **Category references**: F43.2, F45, F44.4 (primary); F82 (secondary)
6. **Association class**: direct

Its evidence status is theoretical: the technique carried *confirmed* until 2026-10-16, when its sources were found not to demonstrate signal injection (Section 4).

6.7 Impact Chains as Associations (Live Dataset)

The live impact-chains dataset (5,546 rows) is built by joins: each technique to every structure sharing one of its addresses, then to every pathway touching that structure, then to that pathway's condition codes. A row therefore records a shared address, not that the technique reaches the structure. The join currently yields rows for 91 of the 173 techniques and 10 structures. A technique with no row has no association recorded, which is not evidence that it has no effect. Because the joins run on addresses, the move to addressing version 2 (Section 3) changes them; the snapshot figures above use version 1.

7 Governance and Neuroethics

This section sets out why a security framework for neural devices needs neuroethics, where neuroethics enters the scoring and the labels, and how the framework relates to policy instruments.

None of it has been examined by an ethicist, clinician or review board.

7.1 Why Neuroethics

Neuroethics asks what may rightly be done to and with a human brain. Roskies divided the field into the ethics of neuroscience, which asks how brain research and technology should be conducted and used, and the neuroscience of ethics [41]; this framework is concerned only with the first.

A security framework needs it because the harms of a compromised neural device are not all harms to systems or data. Consent to a medical device can be partial and can be withdrawn; some effects on tissue or neural function cannot be undone; and a recording of brain activity cannot be reissued. Conventional scoring has no vocabulary for these, and neuroethics supplies one. Ienca and Andorno [22] proposed four rights—cognitive liberty, mental privacy, mental integrity and psychological continuity—and Yuste et al. [53] set out ethical priorities for neurotechnology and artificial intelligence.

The rights are contested. Bublitz argues against inflating the catalogue of rights, while describing Ienca and Andorno's four as better grounded in existing scholarship [8]; Ienca's own later review warns against the same inflation [21]; and mental privacy and mental integrity lack agreed operational definitions [26]. QIF therefore uses the four rights as *labels* for what a technique puts at stake, defines only measurable signal properties, and does not argue that new law is needed.

Neuroethics also limits what a security framework may claim. The project holds eight guardrails as data (`datalake/qif-guardrails.json`), each pairing a caution from the literature with a limit on a component. Among them: activity in a brain region does not identify a cognitive process [36], so TARA catalogues physical interference, not cognitive content; brain data is given more credibility than it has earned [37]; present devices do not read thoughts [24, 51], so every technique separates what is possible now from what is projected; and neuroscience is dual-use, pursued by national security agencies [47], so every threat description is defensive and paired with a control. The evidence rule, the three scoring states and the labelling of diagnostic codes as category references (Sections 4–6) apply these guardrails in practice.

7.2 How Neuroethics Enters the Score

Three commitments are fixed in the NISS arithmetic (Equation 2):

1. **Consent is a scored metric at full weight.** A technique that changes nothing physically still scores if it overrides consent, although it cannot score above 3.0 (Section 5.6). Of the 129 live scored techniques, 68 are rated Explicit, 49 Implicit, 5 Partial and 7 None on CV.
2. **Harm that cannot be undone counts for more.** Reversibility and Neuroplasticity carry full weight, expressing non-maleficence as a number, and the PINS flag applies to any irreversible technique whatever its total.
3. **Reading and disruption are scored separately.** CR and CD carry half weight each, so together they weigh as much as one other metric.

Rights labels are then assigned by rule: High or Critical CR earns mental privacy and cognitive liberty; High or Critical CD or BI earns mental integrity; further labels come from a technique's category, its category references and short hand-made lists. In the live registry 142 of 173 techniques carry at least one label (mental privacy 111, mental integrity 102, cognitive liberty 87, psychological continuity 79). The labels do not feed back into the score.

7.3 Consent Tiers

Each technique's governance record names a consent tier:

Table 11: Consent tiers: snapshot (v1.5) and live counts.

Tier	Requirement	v1.5	Live
Standard	Standard clinical consent	22	42
Enhanced	Extended disclosure and monitoring	51	70
IRB	Full institutional review board / ethics committee review	28	52
Prohibited	Prohibited outside controlled research	2	9
—	Not assigned	6	0

7.4 What an Institutional Programme Has, and QIF Lacks

The guardrails above are self-imposed: no one outside the project checks them. The neuroethics programme of the NIH BRAIN Initiative is a published example of more. It treats neuroethics as concurrent with tool development rather than as review after the fact [38], and rests on four standing activities: a Neuroethics Working Group of neuroethics and neuroscience experts that advises the programme; funded neuroethics research; scanning of the research portfolio for the ethical questions it raises; and workshops on topics that steer the programme [5]. A later account describes this as integrating neuroethics into neuroscience rather than appending it [39].

QIF has none of the first, second or fourth, and performs the third only as the author applying the guardrails to the author's own work. This is a gap statement, not a plan; it keeps the four activities visible as the standard the framework falls short of. Nothing here implies that the NIH, the BRAIN Initiative or its working group has reviewed, endorsed or has any connection to QIF.

7.5 Alignment with Policy Instruments

UNESCO's Recommendation on the Ethics of Neurotechnology [48], adopted in November 2025, sets out values, principles and policy action areas. Table 12 maps its elements to QIF components. The mapping is the author's self-assessment; it has not been reviewed by UNESCO or any third party, and "mapped" means that a component addresses the element, not that the element is met.

Table 12: Self-assessed mapping of UNESCO Recommendation elements to QIF components.

UNESCO Element	Type	Status	QIF Component
Human rights & dignity	Value	Mapped	Rights labels, consent tiers
Health & well-being	Value	Mapped	PINS flag, biological impact scoring
Diversity	Value	Mapped	Open-source, multi-stakeholder model
Sustainability	Value	Mapped	Apache 2.0 license, post-trial access
Professional integrity	Value	Mapped	Transparency audit trail
Proportionality	Principle	Mapped	Graduated severity scoring
Freedom of thought	Principle	Mapped	Cognitive liberty label, consent metric
Privacy	Principle	Mapped	Cognitive reconnaissance (CR) / disruption (CD) metrics
Protection of children	Principle	Mapped	Age-tiered assent framework (governance documents)
Consumer protection	Policy	Mapped	Default-deny architecture (governance documents)
Enhancement regulation	Policy	Partial	Technical infrastructure only
Workplace protections	Policy	Mapped	Mental privacy scoring
Behavioral influence	Policy	Mapped	Consent violation metric (CV)
Health and well-being	Policy	Mapped	NISS scoring, PINS flag
Oversight & governance	Policy	Mapped	Regulatory mapping (below)
Access & equity	Policy	Mapped	Open-source release

The table lists 16 rows. The v1.5 text described them as 15 of 17 elements "implemented" with

two partial, which did not match its own table; "implemented" is replaced by "mapped" because no row has been tested against the Recommendation. Two components cited (the age-tiered assent framework and default-deny architecture) are described in the project's governance documents and are not part of the TARA or NISS data analysed here.

7.6 Regulatory Mapping

The framework relates to existing instruments as follows. None requires TARA or NISS, and no regulator has reviewed either.

- **FDA Section 524B** [50]: mandatory cybersecurity documentation for connected medical devices, including threat modeling; TARA could inform the threat model and NISS could sit beside CVSS in it
- **EU MDR 2017/745** [15]: medical device risk management, for which TARA offers a neural threat taxonomy
- **ISO 14971** [25]: risk management for medical devices; the NIC offers a structured association from technical failure to categories of harm
- **HIPAA** [49] and **GDPR** [14]: health and sensitive personal data, under which neural data may fall (GDPR Article 9)
- **Colorado Privacy Act** [46]: amended in 2024 to classify neural data as sensitive data

8 Case Studies

To show the practical difference between CVSS-only and CVSS+NISS scoring, we score four registry entries with both systems, one from each gap group plus a read-only case.

Correction. In v1.5, Cases 1–4 used identifiers whose registry entries describe different techniques (QIF-T0015, T0034 and T0050 are directed-energy damage, working-memory poisoning and hardware fault injection, not the P300, calibration and surveillance attacks described), and their NISS vectors and scores did not match the registry. Some case scores were also computed as an unweighted mean of six, which is not Equation 2. This revision replaces them with entries as stored in the live registry.

Method and caveat. NISS scores are recomputed from the stored vectors with Equation 2 and match the stored values. CVSS v4.0 base scores were computed from the stored base vectors with the open-source `cvss` Python library (v3.6) and have not been checked against the FIRST.org calculator. The vectors were unchanged between the snapshot and the live registry, but the snapshot file stored NISS scores that do not reproduce from the equation (for example 3.4 for QIF-T0035 against 2.7). None of these techniques has been executed against an implanted device; Case 5 is the only analysis of a real vulnerability.

Table 13: CVSS v4.0 vs. NISS for four registry entries (live registry).

Technique	Name	Status	CVSS	NISS
QIF-T0001	Signal injection	Theoretical	5.8 (Med)	6.1 (Med), PINS
QIF-T0035	P300 interrogation	Demonstrated	6.9 (Med)	2.7 (Low)
QIF-T0058	Calibration poisoning	Demonstrated	7.1 (High)	6.0 (Med)
QIF-T0072	Transducer inversion	Demonstrated	5.6 (Med)	2.0 (Low)

8.1 Case 1: Signal Injection (QIF-T0001, Gap Group 3)

CVSS rates this Medium (5.8): the vector requires physical access and attack preconditions, with high integrity and availability impact on the system and high integrity impact on subsequent systems. What it cannot express is the consequence for the person: current delivered at the electrode-tissue boundary can cause tissue stress or involuntary activation.

`NISS:1.1/BI:H/CR:H/CD:H/CV:E/RV:P/NP:T` gives 6.1 and sets PINS because BI is High. The technique is theoretical: its two listed sources are a position paper and a P300 probing study, so the score describes a modeled impact, not an observed one.

8.2 Case 2: P300 Interrogation (QIF-T0035, Read-Only)

CVSS rates this Medium (6.9) for high confidentiality impact on the device and on subsequent systems. The leaked data is the person's own evoked responses, which reveal recognition of stimuli such as PINs, faces or places [29]. `NISS:1.1/BI:N/CR:H/CD:H/CV:E/RV:F/NP:N` gives $(0 + 0.5 \cdot 6.7 + 0.5 \cdot 6.7 + 6.7 + 0 + 0)/5 = 2.68 \rightarrow 2.7$. The low score is a direct consequence of the read-only ceiling (Section 5.6); the CR and CV values, not the total, carry the information. The registry's CD:H for a read attack is a scoring choice that warrants review.

8.3 Case 3: Calibration Poisoning (QIF-T0058, Gap Group 2)

CVSS rates this High (7.1) for integrity impact during calibration. Corrupted calibration data makes the decoder learn wrong mappings between neural signals and intended actions, and the user may adapt to the corrupted mapping. `NISS:1.1/BI:L/CR:H/CD:H/CV:I/RV:P/NP:T` gives $(3.3 + 3.35 + 3.35 + 10 + 6.7 + 3.3)/5 = 6.0$; CV:I records that the user cannot detect the corruption. The entry's source is Schroder et al. [43]; the registry pairs it with signed calibration data and recalibration audits as controls.

8.4 Case 4: Transducer Inversion (QIF-T0072, Gap Group 1)

A speaker is re-tasked as a microphone. CVSS rates this Medium (5.6), and its confidentiality metrics describe the impact well. NISS adds little: `NISS:1.1/BI:N/CR:N/CD:N/CV:I/RV:F/NP:N` gives 2.0, and the dual-use class is silicon-only. The case is included to show that NISS is not meant to raise every score.

8.5 Case 5: Real-World Vulnerability Disclosure

The QIF framework has been applied to real vulnerability research. During systematic analysis of the BCI software ecosystem, we identified a multi-phase exploit chain in an open-source library used in clinical and research BCI pipelines.

The exploit chain demonstrates escalation from device-side vulnerabilities (S2/S3) to potential impact on clinical decision-making through corrupted data. CVSS scores the software vulnerabilities accurately; NISS records the downstream risk to patients whose clinical care depends on the integrity of the data stream.

Responsible disclosure is in progress. Specific vulnerability details, including affected software and CWE identifiers, will be published after coordinated disclosure concludes.

9 Limitations and Future Work

We present these limitations transparently to guide future validation efforts and to prevent overstatement of the framework's current maturity. QIF, TARA and NISS are proposed and unvalidated: not independently peer-reviewed as a whole, and not adopted by any standards body or device maker.

9.1 No Empirical Validation on Real BCI Devices

The QIF framework has not been validated against operational BCI hardware. The TARA taxonomy was developed through literature review, threat modeling, and systematic analysis rather than penetration testing of actual neural devices. While the framework has been applied

to one real software vulnerability (Section 8), this covers only device-side code. Validation against interface and neural threats requires access to implanted BCI patients and clinical environments—resources unavailable to independent researchers.

9.2 Reproducibility of the v1.5 Figures

For this revision the snapshot figures were recomputed from the registry file committed with v1.5. The evidence-status, filed-severity, category, dual-use, consent-tier, PINS and Neural Impact Chain counts reproduce exactly. Three do not: the gap groups (published 13/31/64, totalling 108 or 99.1%; file 14/30/59, totalling 103 or 94.5%), the NISS severity distribution (published 19/37/52/1 high to none; file 21/37/50/1), and several stored NISS scores, which do not reproduce from Equation 2. The cause was not identified. The text of this revision uses the reproducible gap counts; the published ones are kept in the tables for the record. The case studies and representative techniques of v1.5 did not match the registry at all and were replaced (Sections 4 and 8).

9.3 Diagnostic Category Associations Not Clinically Validated

The Neural Impact Chain associates techniques with diagnostic category references through neuroanatomical addresses and functional neuroscience. These associations have not been reviewed by psychiatrists, neurologists or clinical neuroscientists. They are references for threat modeling, not diagnostic claims, and must not inform clinical decision-making.

9.4 NISS Weights Not Calibrated

The default profile weights BI, CV, RV and NP at 1.0 and CR and CD at 0.5 (Equation 2). Neither these nor the four context profiles have been calibrated against empirical data, expert elicitation or clinical outcomes, and the read-only ceiling of 3.0 (Section 5.6) is a direct consequence of the default choice. Calibration requires expert panel scoring of representative scenarios, sensitivity analysis across weight configurations, and correlation with observed outcomes when available.

9.5 No Interrater Reliability Study

NISS scores in the TARA registry were assigned by a single analyst (the author) with AI assistance. No interrater reliability study has been conducted. CVSS interrater reliability is a known challenge [16]; NISS, with its novel neural-specific metrics, likely faces greater variability. A formal study with domain experts from both cybersecurity and neuroscience is needed.

9.6 Evidence and Completeness

The analysed snapshot contains 109 techniques (v1.5), of which 31 are theoretical and 1 speculative. The live registry holds 173 techniques and 1 variant across 17 tactics (v4.3), of which 95 are theoretical. 103 carry CVSS v4.0 base vectors; of the NISS scores 129 are complete, 10 are provisional ranges and 34 are unscored under the evidence rule. The 129 scored techniques were scored before that rule and have not been re-examined against it, so the scored count is expected to fall. Live figures are not substituted into the analysis, because every proportion in Sections 4–7 has a numerator computed over the snapshot; re-pointing the denominator would produce figures that describe no actual registry state. Current values are derivable from `datalake/qtara-registrar.json` in the project repository. The addressing model has not been reviewed by a human neuroanatomist, and the catalogue records techniques one person at a time, so attacks spread across a population are outside its model (Section 2).

9.7 Single-Author Bias

The framework was developed by a single independent researcher. While multi-model AI review was used throughout development (Claude, Gemini, ChatGPT), the architectural decisions, scoring assignments, and clinical associations reflect a single perspective, and AI review does not substitute for independent human review.

9.8 AI Tool Disclosure

In accordance with arXiv, ACM, and IEEE policies on AI-assisted research, we disclose the following. Large language models (Claude 3.5/4, Gemini 2.0, ChatGPT-4o) were used during the development of this framework and versions up to v1.5. Their roles, by section:

- **Literature review** (Sections 2, 3, 7): AI assisted in identifying relevant publications and summarizing prior work. All citations were independently verified by the author via DOI resolution and publisher page confirmation (see version history below).
- **Code generation** (figures, TARA data pipeline): AI generated Python scripts for data analysis, visualization, and figure generation. All code was reviewed and modified by the author.
- **Writing assistance** (all sections): AI provided draft text that was substantially rewritten, restructured, and verified by the author. For versions up to v1.5, the proportion of AI-generated text retained verbatim is estimated at less than 15%.
- **Cross-validation**: AI models were used to cross-check factual claims across multiple systems (Claude, Gemini, ChatGPT) to reduce single-model bias. Disagreements between models were resolved by the author through primary source verification.

This revision (v1.6). The v1.6 changes were drafted on 2026-10-17 by Claude Opus 5.5 (Anthropic), operating through Claude Code, from the project's data files, the v9 whitepaper and the repository history. The AI system wrote the new and rewritten text (the abstract; Section 1.1; the cognitive-neurosecurity subsection of Section 2; Section 3; the evidence, hazards and representative-technique subsections of Section 4; the gap analysis and Section 5.6; the walkthrough and association subsections of Section 6; Sections 7.1 and the remainder of Section 7; Section 8; this section's reproducibility, evidence and version notes; and Section 10). It recomputed every figure it added from the data files with scripts it ran, computed the CVSS scores in Section 8, and re-resolved each added citation through the Crossref or arXiv API on the same date. Text new in v1.6 is predominantly AI-drafted and was pending the author's review when drafted; it should not be read as author-reviewed until the Zenodo version description says so.

Human-originated contributions: The original band model, the TARA taxonomy structure, the NISS scoring methodology, the Neural Impact Chain method, all diagnostic category associations, all NISS metric assignments, and the research conclusions of v1.0–v1.5 were conceived, designed and decided by the author. The addressing revision, the evidence rule and the retirement of the confirmed status were made in the project's data before this revision, which reports them.

The author takes full responsibility for all content in this paper, irrespective of how it was generated. A complete, auditable transparency log documenting AI contributions and verification steps is maintained at https://github.com/qinnovates/qinnovate/blob/main/governance/TRANSPARENCY.md.

Version history. An earlier version of this preprint (v1.0) contained citation errors introduced during AI-assisted bibliography construction, including three fabricated entries. These were corrected in v1.1 through a two-pass independent verification audit. Version 1.2 corrected an internal percentage inconsistency and added the author responsibility statement above. Version 1.3 expanded the regulatory context with FDORA/PATCH Act Section 524B analysis

and added Schroder et al. (2025) to the related work. Version 1.4 added the Sherman & Guillery (2006) citation, replaced a mislabeled DBS citation with Hallett (2007) for TMS, corrected case study scores, fixed the dual-use classification label, and completed a citation and claims verification pass. Version 1.5 expanded the registry from 102 to 109 techniques, split the NISS Cognitive Integrity metric into Cognitive Reconnaissance and Cognitive Disruption (NISS v1.1), normalized default weights, and added neurorights mapping for 103 of 109 techniques. Version 1.6 (October 2026) replaces the hourglass stack with addressing version 2; adds the evidence rule, the three scoring states, the retirement of the confirmed status, the hazards dataset and NISS's stated limits; replaces case studies and representative techniques that did not match the registry; reports which snapshot figures do and do not reproduce; adds arguments for why neurosecurity and neuroethics are needed, with twenty newly cited sources; qualifies or removes priority claims, universal negatives and diagnostic language; removes six figures whose hard-coded values or labels no longer matched the tables or the model; and condenses the future-work list.

Note on AI use and preprint status: I did not find anything in Zenodo's terms of use that prohibits using AI to help write a preprint, so I used it, heavily, to get this initial version out the door. The AI helped me move faster, but the ideas, the architecture, the scoring, the clinical mappings, those are mine. This preprint is me planting a seed. The real work is what comes next: writing it properly with human collaborators, getting the DSM-5-TR mappings reviewed by actual clinicians, running empirical validation, and revising it until it holds up to scrutiny. There is a lot more to do.

9.9 Future Work

1. **Re-examination of existing scores**: apply the evidence rule to the 129 scored techniques
2. **Expert review**: a neuroanatomist's review of the addressing model before the routes are integrated, and clinical review of the category associations
3. **Interrater reliability and weight calibration**: a formal study with cybersecurity and neuroscience experts, with sensitivity analysis of the weights
4. **Checks in the build**: run the band check, a score recomputation and the route gate in continuous integration
5. **Standards input**: offer NISS to FIRST.org's CVSS community only after at least one independent researcher has reviewed the scoring method; no submission has been made
6. **Empirical testing**: testing against BCI hardware in controlled environments with ethics approval
7. **Exploratory directions** documented in the repository and not claims of this paper, each requiring ethics review before any implementation: hash-chained audit logs for interface firmware; an EEG dataset pipeline for validating components (https://github.com/qinnovates/qinnovate/blob/main/shared/EEG-DATA-PIPELINE.md); a commodity-hardware vision-prosthesis security prototype (https://github.com/qinnovates/qinnovate/blob/main/docs/research/lidar-bci-bridge-technical-spec.md); and consent-gated public-safety concepts

10 Conclusion

Brain-computer interfaces are moving from laboratory prototypes to commercial medical devices. Security frameworks designed for information technology are necessary for these devices but not sufficient: a compromised neural device can act on a person, can expose data that cannot be reissued, and can outlast the party that supports it. That is the case for neurosecurity. Because the harms include violated consent and effects that cannot be undone, and because claims about brains are easily overstated, a neurosecurity framework also needs neuroethics, both to decide what counts as harm and to limit what the framework itself may claim.

This paper presented the QIF framework, a proposed and unvalidated system comprising an addressing model for what a device touches, a threat catalogue (TARA), an impact-only CVSS v4.0 extension (NISS, six metrics, with three scoring states), and the Neural Impact Chain, a threat-modeling method that associates techniques with diagnostic category references. In the 109-technique snapshot analysed here, CVSS cannot express the primary impact of 59 techniques and misses part of it for 30 more; 71.6% of techniques have a recorded therapeutic analog, showing how often the mechanism of an attack is the mechanism of a therapy; and 56.0% are in a direct or indirect association class.

This revision also corrected the paper. It withdrew the hourglass stack and its chokepoint, retired an evidence status that conflated unrelated kinds of evidence, replaced case studies that did not match the registry, and reported which published figures could not be reproduced. The framework's limits remain: no empirical validation on BCI hardware, no clinical review of the category associations, no expert review of the addressing model, and single-author scoring. These are invitations to collaborate, not reasons to delay publication.

The framework, catalogue, NISS specification and scoring data are released under the Apache 2.0 license. We invite review and correction from the neuroscience, neuroethics, cybersecurity and clinical communities; a finding that an entry is wrong is as useful as a finding that it holds.

References

[1] American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision (DSM-5-TR)*. American Psychiatric Association Publishing, 2022. doi: 10.1176/appi.books.9780890425787. Verified 2026-02-16 via Crossref API.

[2] Bryce Allen Bagley and Claudia K Petritsch. A mathematical framework for the problem of security for cognition in neurotechnology. *arXiv preprint arXiv:2403.07945*, 2024. URL https://arxiv.org/abs/2403.07945. Preprint, not peer reviewed. Cited at v4 (2025-01-26). AI-suggested, verified 2026-10-07 via arXiv API (title, authors and year taken from the API record). Catalogue ID C134. Re-resolved 2026-10-17 via arXiv API (ID, authors, title, dates matched).

[3] Bryce-Allen Bagley, Nathaniel Rose, Quintus Kilbourn, and Matthew Canham. Threat vectors and the state of the art in defense methods for security in neurotechnology. *arXiv preprint arXiv:2607.10451*, 2026. URL https://arxiv.org/abs/2607.10451. Preprint, not peer reviewed. AI-suggested, verified 2026-10-07 via arXiv API (title, authors and year taken from the API record). Catalogue ID C135. Re-resolved 2026-10-17 via arXiv API (ID, authors, title, dates matched).

[4] Rebecca Bellan. OpenAI invests in Sam Altman's brain computer interface startup Merge Labs. TechCrunch, January 2026. URL https://techcrunch.com/2026/01/15/openai-invests-in-sam-altmans-brain-computer-interface-startup-merge-labs/. $252M seed round at $850M valuation; co-founded by Sam Altman, Alex Blania, and Caltech researchers. Verified 2026-02-16 via URL check.

[5] Diana W. Bianchi, Judith A. Cooper, Joshua A. Gordon, Jill Heemskerk, Richard Hodes, George F. Koob, Walter J. Koroshetz, David Shurtleff, Paul A. Sieving, Nora D. Volkow, James D. Churchill, and Khara M. Ramos. Neuroethics for the national institutes of health brain initiative. *The Journal of Neuroscience*, 38(50):10583–10585, 2018. doi: 10.1523/JNEUROSCI.2091-18.2018. Verified 2026-10-16 via Crossref. Catalogue ID I22. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[6] Tamara Bonaci, Ryan Calo, and Howard J. Chizeck. App stores for the brain: Privacy & security in brain-computer interfaces. *IEEE Technology and Society Magazine*, 34(2):32–39, 2015. doi: 10.1109/MTS.2015.2425551. Verified 2026-02-16 via Crossref API.

[7] Clémentine Bosch-Bouju, Brian I. Hyland, and Louise C. Parr-Brownlie. Motor thalamus integration of cortical, cerebellar and basal ganglia information: implications for normal and parkinsonian conditions. *Frontiers in Computational Neuroscience*, 7:163, 2013. doi: 10.3389/fncom.2013.00163. Verified 2026-10-09 via Crossref. Article number from the Europe PMC record. Catalogue ID N75. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[8] Jan Christoph Bublitz. Novel neurorights: From nonsense to substance. *Neuroethics*, 15(1):7, 2022. doi: 10.1007/s12152-022-09481-3. Verified 2026-10-17 via Crossref API (DOI, author, title, journal, volume, article number and year matched; author given name corrected to Jan Christoph from the Crossref record).

[9] Carmen Camara, Pedro Peris-Lopez, and Juan E. Tapiador. Security and privacy issues in implantable medical devices: A comprehensive survey. *Journal of Biomedical Informatics*, 55:272–289, 2015. doi: 10.1016/j.jbi.2015.04.007. Verified 2026-02-16 via Crossref API.

[10] Emmanuelle Courtiol and Donald A. Wilson. The olfactory thalamus: unanswered questions about the role of the mediodorsal thalamic nucleus in olfaction. *Frontiers in Neural Circuits*, 9:49, 2015. doi: 10.3389/fncir.2015.00049. Verified 2026-10-09 via Crossref. Article number from the Europe PMC record. Catalogue ID N76. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[11] Stephen Deering. Watching the waist of the protocol hourglass. Keynote, IETF 51 Plenary, 2001. URL https://www.ietf.org/proceedings/51/slides/plenary-1/. Originally presented at ICNP '98. Foundational description of the Internet hourglass architecture. Verified 2026-02-16 via URL check.

[12] Tamara Denning, Yoky Matsuoka, and Tadayoshi Kohno. Neurosecurity: Security and privacy for neural devices. *Neurosurgical Focus*, 27(1):E7, 2009. doi: 10.3171/2009.4.FOCUS0985. Introduces the term "neurosecurity". Verified 2026-02-16 via Crossref API.

[13] Brian L. Edlow, Emi Takahashi, Ona Wu, Thomas Benner, Guangping Dai, Lihong Bu, Patricia Ellen Grant, David M. Greer, Steven M. Greenberg, Hannah C. Kinney, and Rebecca D. Folkerth. Neuroanatomic connectivity of the human ascending arousal system critical to consciousness and its disorders. *Journal of Neuropathology & Experimental Neurology*, 71(6):531–546, 2012. doi: 10.1097/nen.0b013e3182588293. Verified 2026-10-09 via Crossref. Catalogue ID N74. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[14] European Parliament and Council. General data protection regulation (GDPR) — regulation (EU) 2016/679, 2016. URL https://gdpr-info.eu/. Verified 2026-02-16 via URL check.

[15] European Parliament and Council. Regulation (EU) 2017/745 on medical devices (MDR), 2017. URL https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32017R0745. Verified 2026-02-16 via URL check.

[16] FIRST.org. Common vulnerability scoring system v4.0 specification, 2023. URL https://www.first.org/cvss/v4-0/cvss-v40-specification.pdf. Verified 2026-02-16 via URL check.

[17] FIRST.org. CVSS v4.0 user guide, 2023. URL https://www.first.org/cvss/v4.0/user-guide. Section 3.11: Extension Framework. Verified 2026-02-16 via URL check.

[18] Mark Hallett. Transcranial magnetic stimulation: A primer. *Neuron*, 55(2):187–199, 2007. doi: 10.1016/j.neuron.2007.06.026. Verified 2026-02-18 via PubMed PMID:17640522.

[19] Daniel Halperin, Thomas S. Heydt-Benjamin, Benjamin Ransford, et al. Pacemakers and implantable cardiac defibrillators: Software radio attacks and zero-power defenses. In *Proceedings of the 2008 IEEE Symposium on Security and Privacy*, pages 129–142, 2008. doi: 10.1109/SP.2008.31. Verified 2026-02-16 via Crossref API.

[20] Nathan Higgins, Brette Blakely, Roland Everingham, Frederic Gilbert, Sarah Griffin, Alexander R. Harris, Sally Herring, Calvin Wai Loon Ho, Kate Hoy, Scott Kiel-Chisholm, Julian Koplin, Sharon Lawn, Allan McCay, Nitya Phillipson, Bernadette Richards, Jeffrey V. Rosenfeld, Ehsan Shamsi Gooshki, John Noel Viana, John Gardner, and Adrian Carter. Recommendations on post-trial responsibility in implantable neural device research: a multidisciplinary consensus study. *BMC Medical Ethics*, 27(1), 2026. doi: 10.1186/s12910-026-01475-7. Verified 2026-10-06 via Crossref API (title, first author and year matched). Catalogue ID GP25. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[21] Marcello Ienca. On neurorights. *Frontiers in Human Neuroscience*, 15:701258, 2021. doi: 10.3389/fnhum.2021.701258. Synthesis and critique of both Ienca/Andorno and Yuste neurorights frameworks. Verified 2026-03-04 via DOI. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[22] Marcello Ienca and Roberto Andorno. Towards new human rights in the age of neuroscience and neurotechnology. *Life Sciences, Society and Policy*, 13(1):5, 2017. doi: 10.1186/s40504-017-0050-1. Verified 2026-02-16 via Crossref API.

[23] Marcello Ienca and Pim Haselager. Hacking the brain: Brain-computer interfacing technology and the ethics of neurosecurity. *Ethics and Information Technology*, 18(2):117–129, 2016. doi: 10.1007/s10676-016-9398-9. Verified 2026-02-16 via Crossref API.

[24] Marcello Ienca, Pim Haselager, and Ezekiel J Emanuel. Brain leaks and consumer neurotechnology. *Nature Biotechnology*, 36(9):805–810, 2018. doi: 10.1038/nbt.4240. Verified 2026-10-07 via Crossref. Catalogue ID E17. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[25] International Organization for Standardization. ISO 14971:2019 — medical devices — application of risk management to medical devices, 2019. URL https://www.iso.org/standard/72704.html. Verified 2026-02-16 via ISO catalog.

[26] Philipp Kellmeyer. 'neurorights'. In *The Cambridge Handbook of Responsible Artificial Intelligence*, pages 412–426. Cambridge University Press, 2022. doi: 10.1017/9781009207898.032. Verified 2026-10-09 via Crossref. Catalogue ID C93. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[27] Andres M. Lozano, Nir Lipsman, Hagai Bergman, et al. Deep brain stimulation: Current challenges and future directions. *Nature Reviews Neurology*, 15:148–160, 2019. doi: 10.1038/s41582-018-0128-2. Verified 2026-02-16 via Crossref API.

[28] Gabriel Lázaro-Muñoz, Michelle T. Pham, Katrina A. Muñoz, et al. Post-trial access in implanted neural device research: Device maintenance, abandonment, and cost. *Brain Stimulation*, 15(5):1029–1036, 2022. doi: 10.1016/j.brs.2022.07.051. Verified 2026-02-16 via Crossref API.

[29] Ivan Martinovic, Doug Davies, Mario Frank, Daniele Perito, Tomas Ros, and Dawn Song. On the feasibility of side-channel attacks with brain-computer interfaces. In *Proceedings of the 21st USENIX Security Symposium*, pages 143–158. USENIX Association, 2012. URL https://www.usenix.org/conference/usenixsecurity12/technical-sessions/presentation/martinovic. Verified 2026-02-16 via URL check.

[30] Lubin Meng, Xue Jiang, and Dongrui Wu. Adversarial robustness benchmark for EEG-based brain-computer interfaces. *Future Generation Computer Systems*, 143:231–247, 2023. doi: 10.1016/j.future.2023.01.028. Verified 2026-02-18 via Crossref API; volume/pages added.

[31] MITRE Corporation. ATT&CK® framework, 2024. URL https://attack.mitre.org/. Verified 2026-02-16 via URL check.

[32] Elon Musk and Neuralink. An integrated brain-machine interface platform with thousands of channels. *Journal of Medical Internet Research*, 21(10):e16194, 2019. doi: 10.2196/16194. Verified 2026-02-16 via Crossref API.

[33] Katrina A. Muñoz, Kristin Kostick, Clarissa Sanchez, Lavina Kalwani, Laura Torgerson, Rebecca Hsu, Demetrio Sierra-Mercado, Jill O. Robinson, Simon Outram, Barbara A. Koenig, Stacey Pereira, Amy McGuire, Peter Zuk, and Gabriel Lázaro-Muñoz. Researcher perspectives on ethical considerations in adaptive deep brain stimulation trials. *Frontiers in Human Neuroscience*, 14:578695, 2020. doi: 10.3389/fnhum.2020.578695. Verified 2026-02-16 via Crossref API.

[34] OECD. Recommendation on responsible innovation in neurotechnology. Adopted by the OECD Council on 11 December 2019, 2019. URL https://legalinstruments.oecd.org/api/print?ids=658&Lang=en. Nine-principle innovation governance framework for neurotechnology. Verified 2026-02-16 and 2026-03-04 via URL. Merges a duplicate entry of the same key that carried the oecd.org/sti landing page; the legal-instruments URL is the canonical text.

[35] Thomas J. Oxley, Peter E. Yoo, Gill S. Rind, et al. Motor neuroprosthesis implanted with neurointerventional surgery improves capacity for activities of daily living tasks in severe paralysis. *Journal of NeuroInterventional Surgery*, 13:102–108, 2021. doi: 10.1136/neurintsurg-2020-016862. Verified 2026-02-16 via Crossref API.

[36] Russell A. Poldrack. Can cognitive processes be inferred from neuroimaging data? *Trends in Cognitive Sciences*, 10(2):59–63, 2006. doi: 10.1016/j.tics.2005.12.004. Verified 2026-10-17 via Crossref API (DOI, author, title, journal, volume, issue, pages and year matched). Reverse inference fallacy.

[37] Eric Racine, Ofek Bar-Ilan, and Judy Illes. fMRI in the public eye. *Nature Reviews Neuroscience*, 6:159–164, 2005. doi: 10.1038/nrn1609. Verified 2026-10-17 via Crossref API (DOI, authors, title, journal, volume, pages and year matched). Neurorealism triad.

[38] Khara M. Ramos, Karen S. Rommelfanger, Henry T. Greely, and Walter J. Koroshetz. Neuroethics and the NIH BRAIN initiative. *Journal of Responsible Innovation*, 5(1):122–130, 2017. doi: 10.1080/23299460.2017.1319035. Verified 2026-10-16 via Crossref. Catalogue ID I23. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[39] Khara M. Ramos, Christine Grady, Henry T. Greely, Winston Chiong, James Eberwine, Nita A. Farahany, L. Syd M. Johnson, Bradley T. Hyman, Steven E. Hyman, Karen S. Rommelfanger, Elba E. Serrano, James D. Churchill, Joshua A. Gordon, and Walter J. Koroshetz. The NIH BRAIN initiative: Integrating neuroethics and neuroscience. *Neuron*, 101(3):394–398, 2019. doi: 10.1016/j.neuron.2019.01.024. Verified 2026-10-16 via Crossref. Catalogue ID I8, whose earlier title and author attribution this entry corrects. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[40] Republic of Chile. Constitutional amendment on neurorights (article 19, no. 1), 2021. URL https://www.bcn.cl/leychile/navegar?idNorma=1166983. Ley 21.383. Constitutional amendment protecting brain activity and information derived from it. Verified 2026-02-16 via URL check.

[41] Adina Roskies. Neuroethics for the new millenium. *Neuron*, 35(1):21–23, 2002. doi: 10.1016/S0896-6273(02)00763-8. Verified 2026-10-07 via Crossref. Catalogue ID E14. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[42] Michael Rushanan, Aviel D. Rubin, Denis Foo Kune, and Colleen M. Swanson. SoK: Security and privacy in implantable medical devices and body area networks. In *Proceedings of the 2014 IEEE Symposium on Security and Privacy*, pages 524–539, 2014. doi: 10.1109/SP.2014.40. Verified 2026-02-16 via Crossref API.

[43] Tyler Schroder, Renée Sirbu, Sohee Park, Jessica Morley, Sam Street, and Luciano Floridi. Cyber risks to next-gen brain-computer interfaces: Analysis and recommendations. *arXiv preprint arXiv:2508.12571*, 2025. URL https://arxiv.org/abs/2508.12571. Also published in Neuroethics 18 (2025). Verified 2026-02-16 via arxiv.org and Springer.

[44] S Murray Sherman. Thalamus plays a central role in ongoing cortical functioning. *Nature Neuroscience*, 19(4):533–541, 2016. doi: 10.1038/nn.4269. Verified 2026-10-09 via Crossref. Catalogue ID N73. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[45] S. Murray Sherman and R. W. Guillery. *Exploring the Thalamus and Its Role in Cortical Function*. MIT Press, 2nd edition, 2006. ISBN 978-0-262-51344-9. URL https://mitpress.mit.edu/9780262513449/exploring-the-thalamus-and-its-role-in-cortical-function/. Verified 2026-02-18 via MIT Press catalog.

[46] State of Colorado. Colorado privacy act — amendment on biological and neural data, 2024. URL https://leg.colorado.gov/bills/hb24-1058. HB 24-1058. Classifies neural data as sensitive data. Verified 2026-02-16 via URL check.

[47] Michael N. Tennison and Jonathan D. Moreno. Neuroscience, ethics, and national security: The state of the art. *PLoS Biology*, 10(3):e1001289, 2012. doi: 10.1371/journal.pbio.1001289. Verified 2026-10-17 via Crossref API (DOI, authors, title, journal, volume, issue, article number and year matched). Dual-use neurotech security warning.

[48] UNESCO. Recommendation on the ethics of neurotechnology. Adopted at the 43rd session of the General Conference, Samarkand, 2025. URL https://www.unesco.org/en/ethics-neurotech/recommendation. Verified 2026-02-16 via URL check.

[49] U.S. Congress. Health insurance portability and accountability act (HIPAA), 1996. URL https://www.govinfo.gov/link/plaw/104/public/191. Verified 2026-02-16 via URL check.

[50] U.S. Congress. Section 524B of the FD&C act — ensuring cybersecurity of devices, 2022. URL https://www.congress.gov/bill/117th-congress/house-bill/2617/text. Added by Section 3305 of FDORA, Division FF of the Consolidated Appropriations Act, 2023 (Pub. L. 117-328). Effective March 29, 2023; RTA enforcement from October 1, 2023. Verified 2026-02-16 via Federal Register.

[51] Anna Wexler. Separating neuroethics from neurohype. *Nature Biotechnology*, 37(9):988–990, 2019. doi: 10.1038/s41587-019-0230-z. Verified 2026-10-07 via Crossref. Catalogue ID E18. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[52] Francis R. Willett, Erin M. Kunz, Chaofei Fan, et al. A high-performance speech neuroprosthesis. *Nature*, 620:1031–1036, 2023. doi: 10.1038/s41586-023-06377-x. Verified 2026-02-16 via Crossref API.

[53] Rafael Yuste, Sara Goering, Blaise Agüera y Arcas, et al. Four ethical priorities for neurotechnologies and AI. *Nature*, 551(7679):159–163, 2017. doi: 10.1038/551159a. Verified 2026-02-16 via Crossref API.

[54] Yue Zhao and Yuan Lin. Consent in flux: when brain-computer interface embodiment undermines informed consent. *BMC Medical Ethics*, 27(1), 2026. doi: 10.1186/s12910-026-01540-1. Verified 2026-10-06 via Crossref API (title, first author and year matched). Catalogue ID GP24. Re-resolved 2026-10-17 via Crossref API (DOI, authors, title, venue, year matched).

[55] Hubert Zimmermann. OSI reference model — the ISO model of architecture for open systems interconnection. *IEEE Transactions on Communications*, 28(4):425–432, 1980. doi: 10.1109/TCOM.1980.1094702. Verified 2026-02-16 via Crossref API.